

DISTRIBUTED SYSTEMS

Project 2

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GOAL

Extend the distributed message delivery system developed in project 1 with:

1. Security;
2. Interaction with External services;
3. Fault tolerance - you will implement fault tolerance of the Messages server.

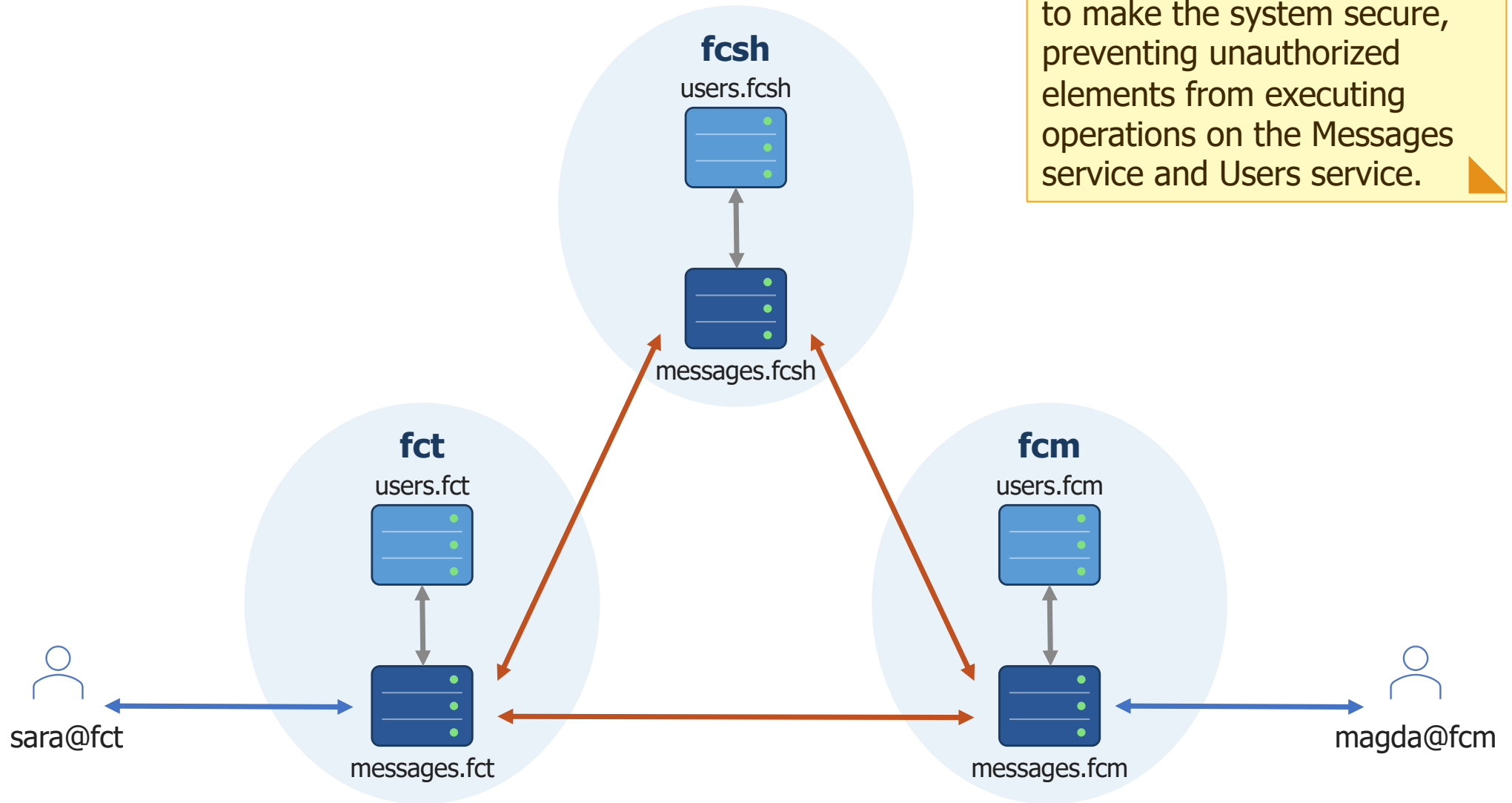
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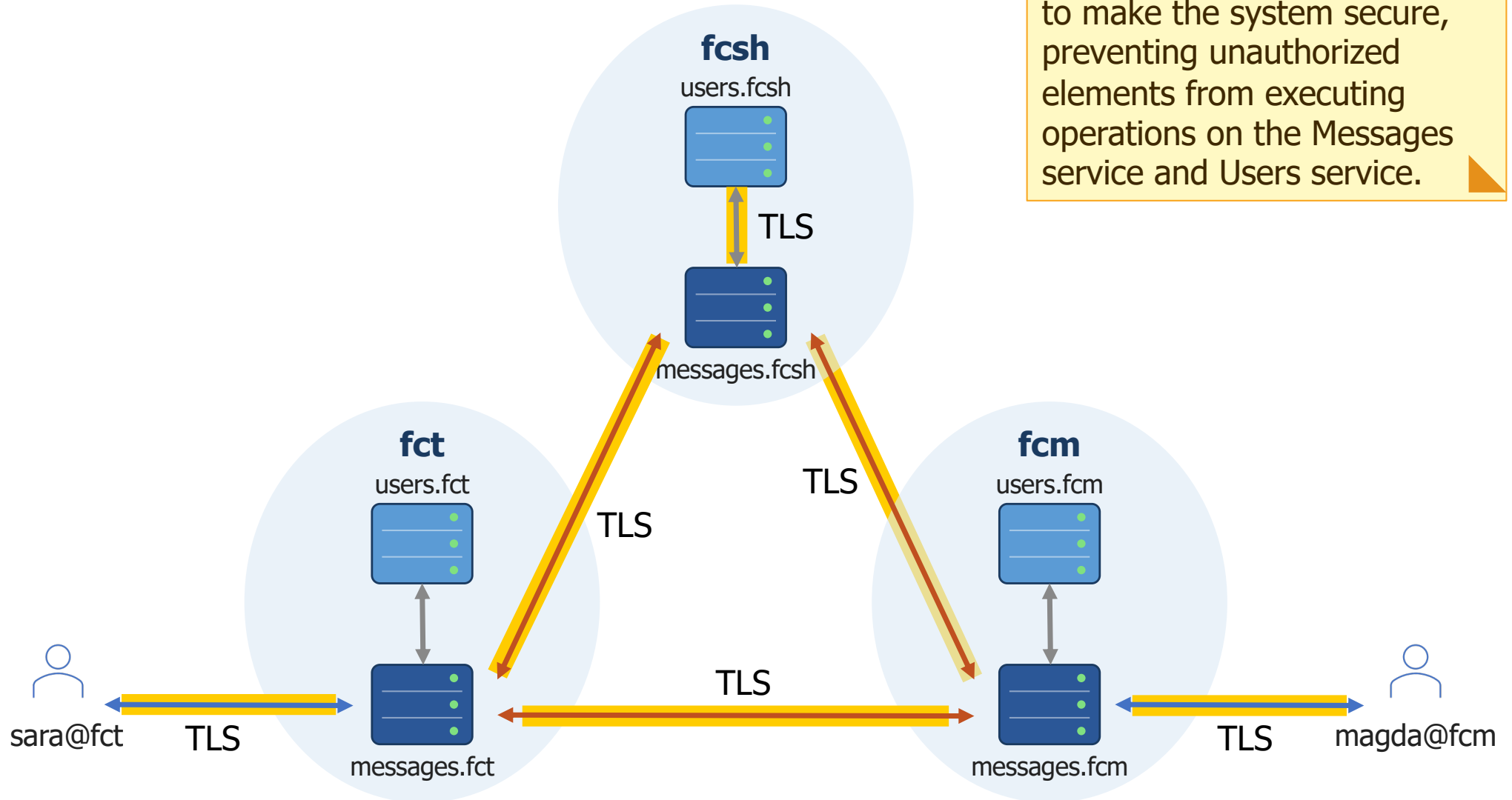
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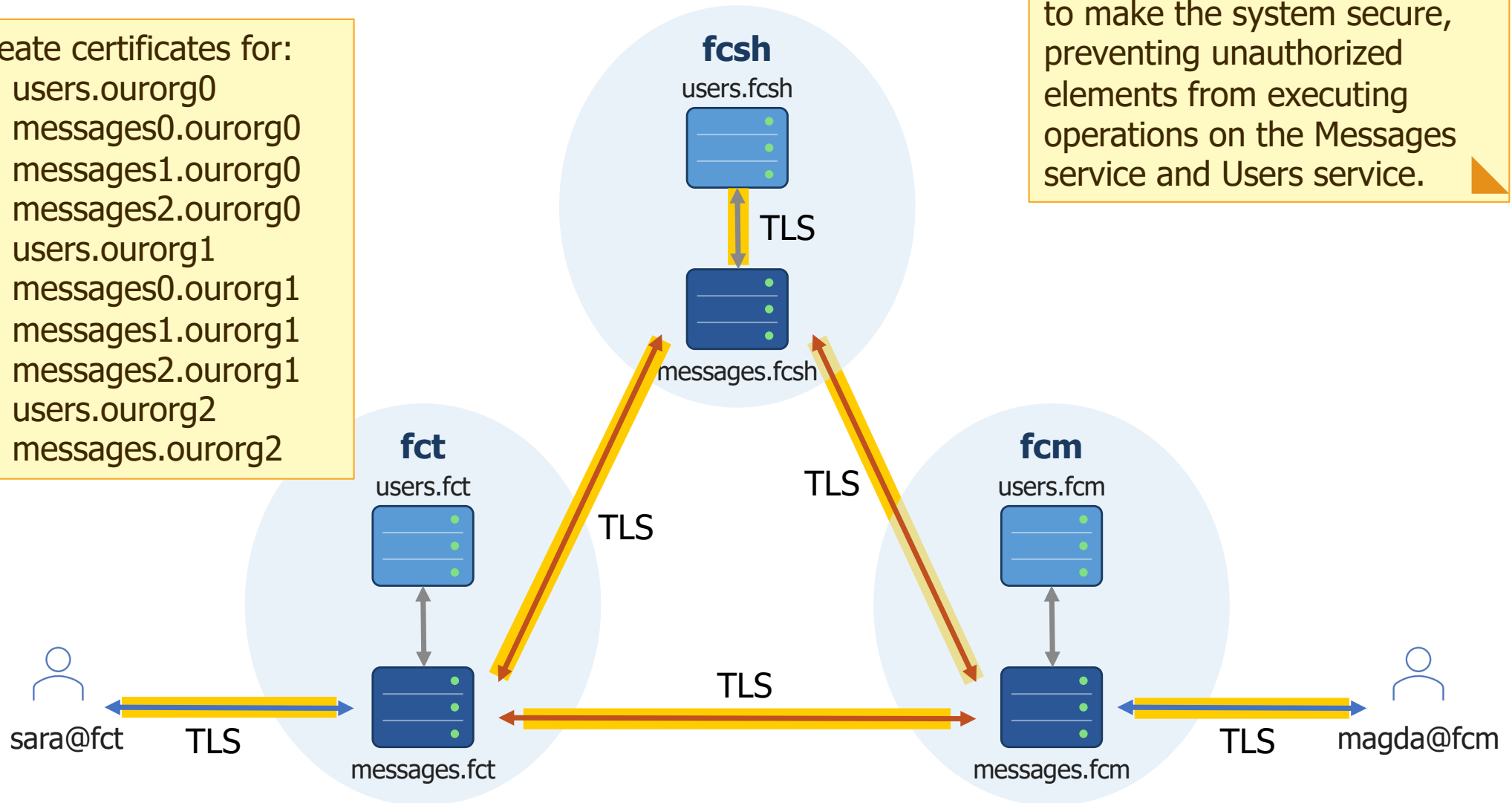
- TLS is all communications – check Lab 8.
- Secure all endpoints – use secret for authenticating servers.

GOAL: SECURITY

Create certificates for:

- users.ourorg0
- messages0.ourorg0
- messages1.ourorg0
- messages2.ourorg0
- users.ourorg1
- messages0.ourorg1
- messages1.ourorg1
- messages2.ourorg1
- users.ourorg2
- messages.ourorg2

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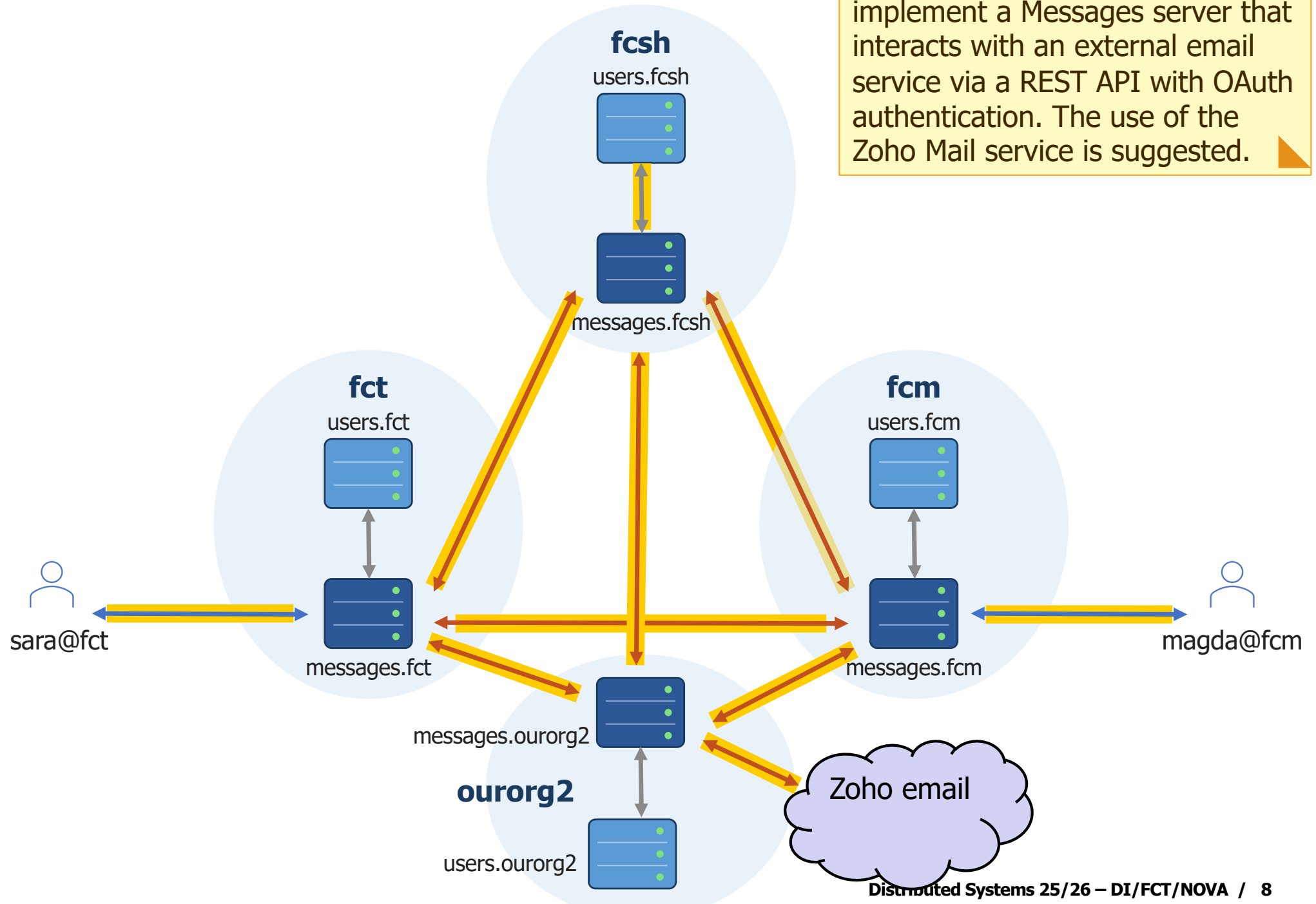
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GOAL: EXTERNAL SERVICE

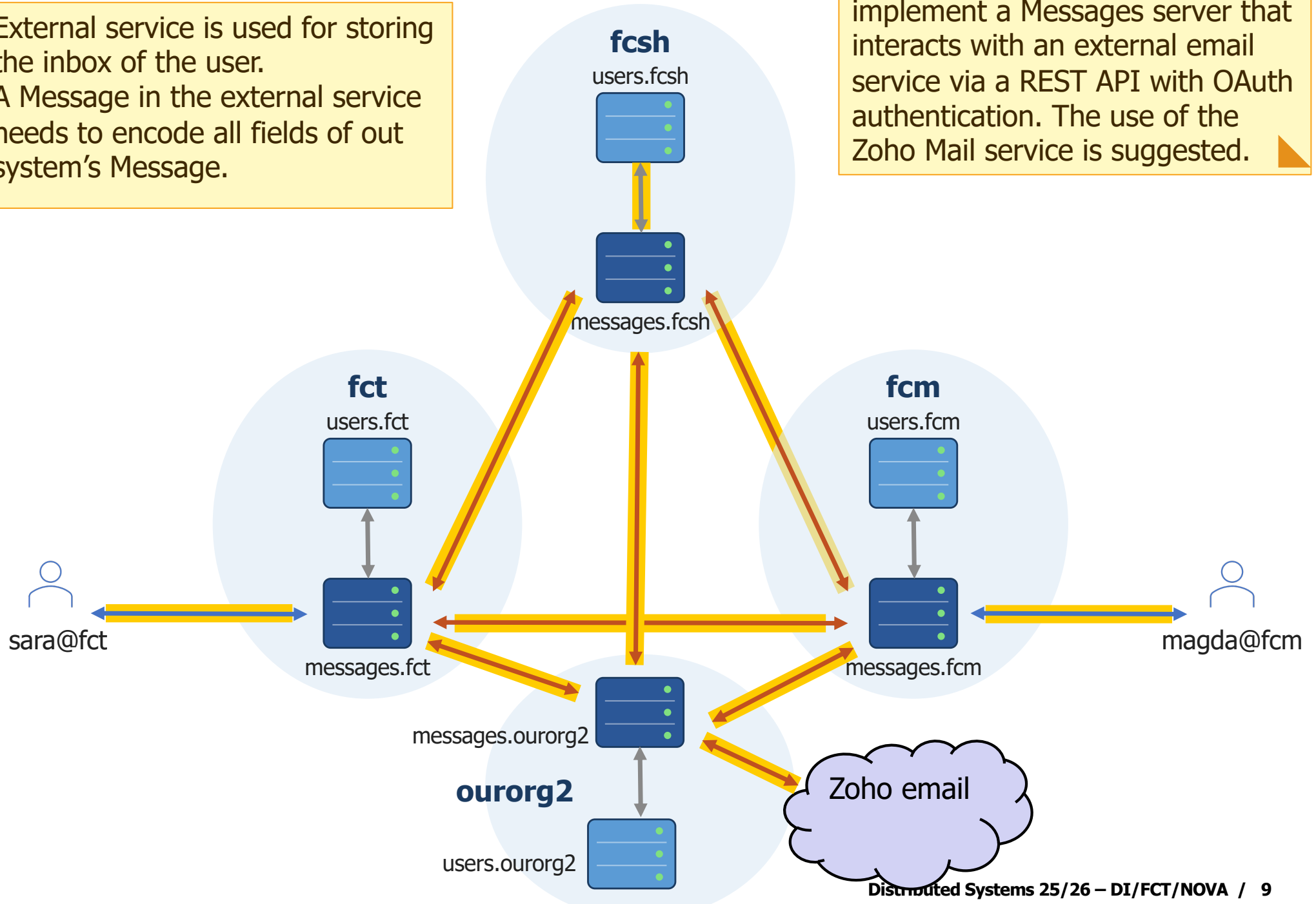
The objective of this feature is to implement a Messages server that interacts with an external email service via a REST API with OAuth authentication. The use of the Zoho Mail service is suggested.



GOAL: EXTERNAL SERVICE

- External service is used for storing the inbox of the user.
A Message in the external service needs to encode all fields of our system's Message.

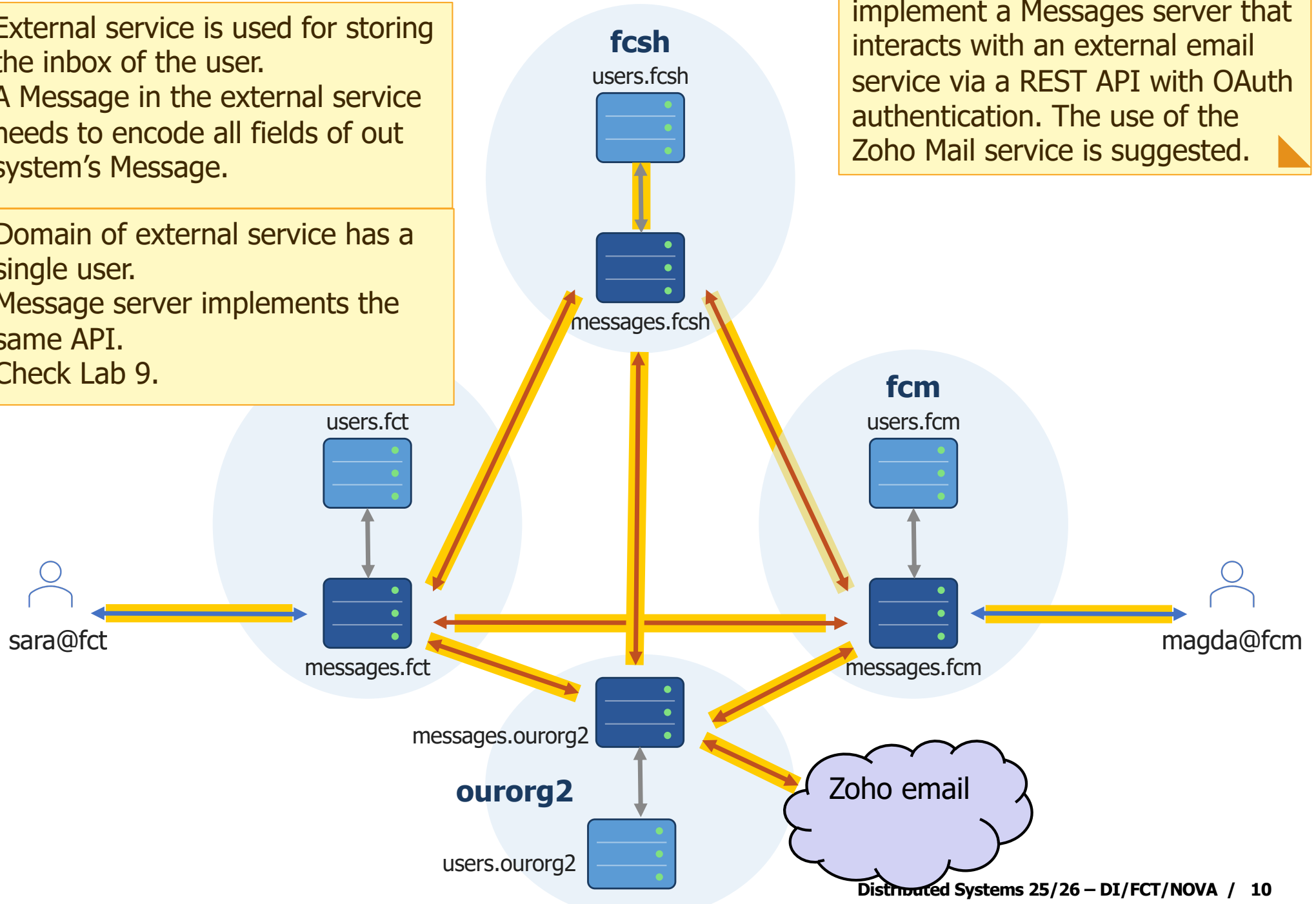
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GOAL: EXTERNAL SERVICE

- External service is used for storing the inbox of the user.
A Message in the external service needs to encode all fields of our system's Message.
- Domain of external service has a single user.
- Message server implements the same API.
- Check Lab 9.

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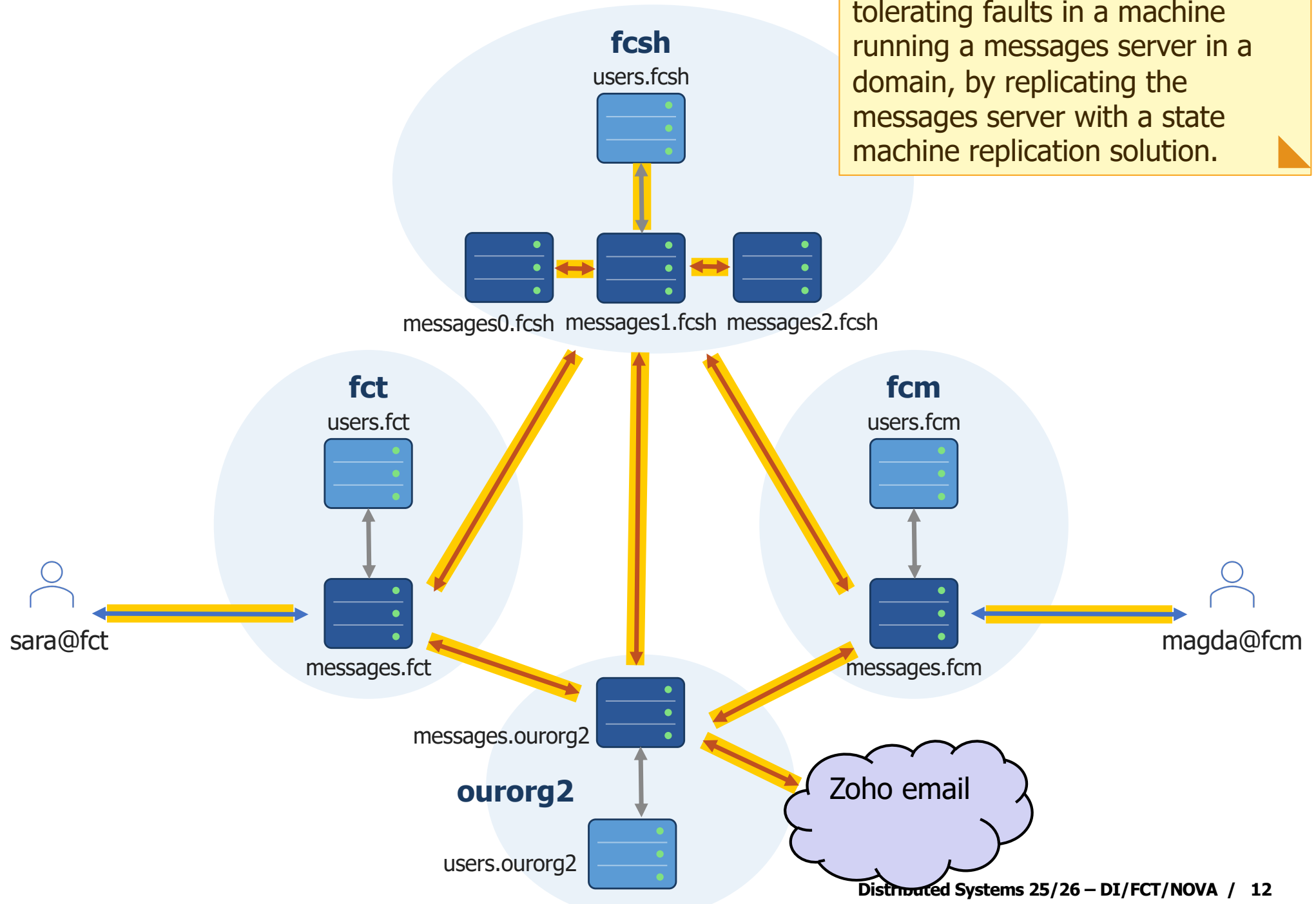
GOAL

Extend the distributed message delivery system developed in project 1 with:

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GOAL: REPLICATION

Implement a solution that allows tolerating faults in a machine running a messages server in a domain, by replicating the messages server with a state machine replication solution.

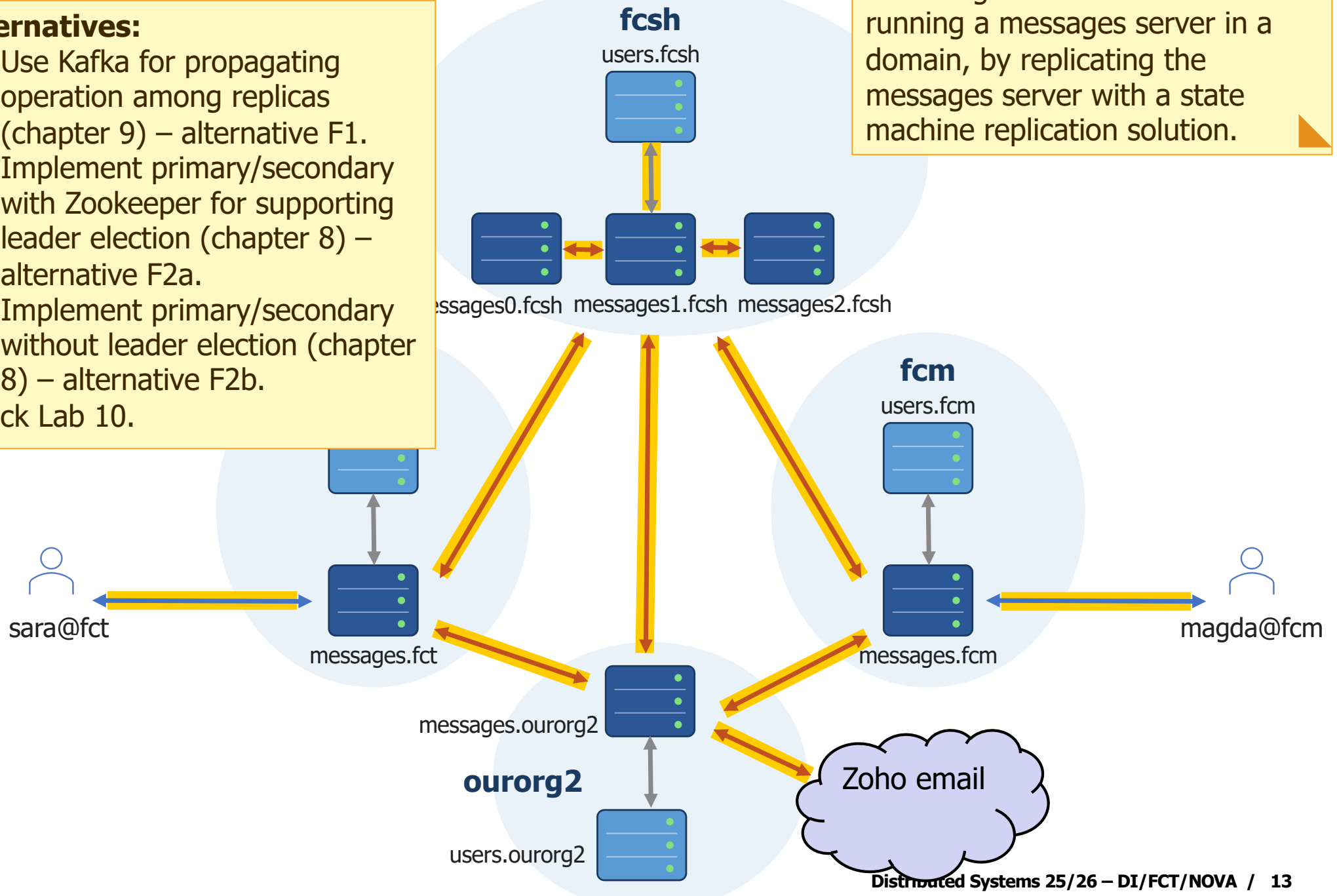


GOAL: REPLICATION

Alternatives:

1. Use Kafka for propagating operation among replicas (chapter 9) – alternative F1.
 2. Implement primary/secondary with Zookeeper for supporting leader election (chapter 8) – alternative F2a.
 3. Implement primary/secondary without leader election (chapter 8) – alternative F2b.
- Check Lab 10.

Implement a solution that allows tolerating faults in a machine running a messages server in a domain, by replicating the messages server with a state machine replication solution.



MORE INFORMATION

You can use your implementation of the first project or use one of the two provided solutions:

- <https://github.com/preguica/sd2526.trab1sol.v1>
- <https://github.com/preguica/sd2526.trab1sol.v2>

A test suite will be provided, as in the first project.

Submission deadline: 26th May, 23:59.